MIL-C-43487A

30 July 1973
SUPERSEDING
MIL-C-43487
10 March 1967

MILITARY SPECIFICATION

CLOTH, TAFFETA, POLYESTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of a polyester fiber cloth.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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PUBLICATION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. The cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Yarn. The polyester yarn for the warp and filling shall be semi-dull, multifilament, polyethylene glycol terephthalate.

3.3 Color. The color of the cloth shall be Tan 448 and shall match the standard sample (see 6.3 and 6.4).

3.3.1 Matching. The dyed and finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 Colorfastness. The dyed and finished cloth shall show fastness to light, laundering, perspiration and crocking equal to or better than the standard sample. When no standard sample is available, the dyed cloth shall show "good" fastness to light, laundering and perspiration, and shall show a Munsell Value for crocking not lower than 8.5 when tested as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.4.

TABLE I. Physical requirements

Weight ounces per sq. yd. (minimum)	Yarns per inch (minimum)		Breaking strength pounds (minimum)		Air permeability (maximum)	Weave
	Warp	Filling	Warp	Filling		
2.8	74	62	155	135	20 cu. feet per minute per square foot	Plain

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3.5 Width. The minimum width inclusive of the selvages of the finished cloth shall be as specified (see 6.2).

* 3.6 Antistatic finish. The cloth shall be given a durable antistatic finish (see 6.5) so that the maximum resistivity of any one sample before laundering shall be 1.0 x 10^{11} ohms per square and the maximum resistivity of any one sample after five launderings shall be 5.0 x 10^{11} ohms per square when tested as specified in 4.4. Only those chemical treatments already approved by the appropriate medical service and so listed herein (see 6.5) or in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

3.7 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.4.

3.8 Shrinkage. The cloth shall not shrink more than 2.0 percent in either the direction of the warp or of the filling when tested as specified in 4.4.

3.9 Length and put-up. Unless otherwise specified, (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up on full width rolls as specified in PPP-P-1133.

3.10 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.11 Marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

3.12 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.2.2 Examination of the end item. Defects found during examination shall be classified in accordance with 4.2.2.1 through 4.2.2.4.

* 4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardzge in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet), shall be scored and assigned demerit points as listed in 4.2.2.1.1, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scales (see 6.6), D for slubs and C for knots, shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 25.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 38.0 points.

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If one roll exceeds the point level, a second sample of the same number of rolls as in the first sample shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds the point level. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times 750} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

NOTE: The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Objectionable odor.
Width of cloth less than specified.
Poor dye penetration, mottled, streaky or cloudy.

4.2.2.2 Examination for length.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of lengths marked on the ticket.

4.2.2.3 Examination for shade. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.2.4 Examination for compliance with the Textile Fiber Products Identification Act and face marking. During the yard-by-yard examination, each

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roll in the sample shall be examined for compliance with the Textile Fiber Products Identification Act and face marking. Each roll not labeled in accordance with the act or with the face marking missing from either or both ends, or stamped on the wrong side shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisons of PPP-P-1133 to determine that packaging, packing and marking comply with the section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 5 yards full width of the finished cloth. The sample size (number of sample units) shall be as shown in the table below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

* TABLE II. Test methods

Characteristic	Requirement paragraph	Test method
Polyester yarn identification	3.2.1	1600 1/
Luster	3.2.1	1/
Colorfastness to:		
Light	3.3.2	5660
Laundering	3.3.2	5614 2/
Perspiration	3.3.2	5680
Crocking	3.3.2	5651

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* TABLE II. Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Weight oz./sq.yd.	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Air permeability	3.4	5450
Weave	3.4	Visual 3/
Antistatic finish:		
Before laundering	3.6	5930 4/
After 5 launderings	3.6	5556 5/ and 5930 4/
pH	3.7	2811
Shrinkage	3.8	5556

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ Only the stain on the nylon fibers of the color transfer cloth shall be evaluated.

3/ One determination per sample unit and the results reported as "pass" or "fail".

4/ Average of the three determinations to nearest 1.0×10^{11} ohms per square.

5/ Cotton laundering procedure.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

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5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of coveralls for wear under controlled laboratory conditions in a dust-proof atmosphere.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Width required (see 3.5).
(c) Length required when other than specified (see 3.9).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Color. A suggested but not mandatory dyestuff formulation for Tan 448 is as follows:

The standard shade sample has been dyed with the following formulation:

(1) Dispersed Yellow 33
(2) Dispersed Red 88
(3) Dispersed Blue FLW

* 6.5 The requirements for the durable antistatic treatment can be met with Aston 123 Refined Onyx Division of Millmaster Onyx Corp., 624 Schuyler Ave., Lyndhurst, N.C. 07071 and Stanax (Not Stanax 1166), Standard Chemical Products Co., Inc., 1301 Jefferson St., Hoboken, N.J. 07030. Other products considered for this use must have the prior approval of the contracting officer.

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* 6.6 Sears Fabric Defect Scales are available from Sears Roebuck and Company, Department 769, "Fabric Defect Replica Kit", 925 South Homan Avenue, Chicago, Illinois 60607.

6.7 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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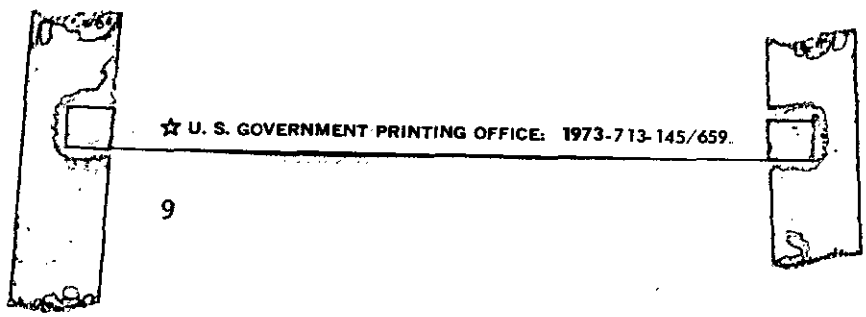
User activities:

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

CLOTH, TAFFETA, POLYESTER MIL-C-43487A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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